PSTAT 5A | Quiz 5 Solutions **Time: 10 minutes**

Problem. A random sample of attendees at a rock concert had the following ages:

$$18,\ 31,\ 25,\ 20,\ 29,\ 33,\ 44,\ 22,\ 20,\ 28.$$

a) Calculate the sample mean.

$$\bar{x} = \frac{18+31+25+20+29+33+44+22+20+28}{10} = \frac{270}{10} = \boxed{27}.$$

b) Find the mode(s).
The value 20 appears twice; all other values appear once. Hence the mode is $\boxed{20}$.

c) Find the five-number summary. Show the ordered data.
Ordered data:

$$18,\ 20,\ 20,\ 22,\ 25,\ 28,\ 29,\ 31,\ 33,\ 44.$$

The median is the average of the fifth and sixth observations:

$$Q_2 = \frac{25+28}{2} = 26.5.$$

The lower half is $18, 20, 20, 22, 25$, so $Q_1 = 20$. The upper half is $28, 29, 31, 33, 44$, so $Q_3 = 31$.

$$\boxed{(\min, Q_1, Q_2, Q_3, \max) = (18, 20, 26.5, 31, 44)}.$$

d) Calculate the interquartile range (IQR).

$$\text{IQR} = Q_3 - Q_1 = 31 - 20 = \boxed{11}.$$